

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk:

Product Description:

Cat No. :

Molecular Formula

Nickel Iron Copper Molybdenum foil, 0.125mm (0.005in) thick, annealed

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43399

Ni:Fe:Cu:Mo; 77:14:5:4 wt%

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address

Enquiry.my@thermofisher.com

Emergency Telephone Number

Tel: +03-5525 7888

CHEMTREC Malaysia 1-800-815-308 (Malay)

CHEMTREC Malaysia (Kuala Lumpur) +(60)-327884561 (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                   |
|------------------------------------------------------|-------------------|
| Skin Sensitization                                   | Category 1 (H317) |
| Carcinogenicity                                      | Category 2 (H351) |
| Specific target organ toxicity - (repeated exposure) | Category 1 (H372) |

**Label Elements**


Signal Word

Danger

**Hazard Statements**

H317 - May cause an allergic skin reaction

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H351 - Suspected of causing cancer

H372 - Causes damage to organs through prolonged or repeated exposure

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P272 - Contaminated work clothing should not be allowed out of the workplace

P280 - Wear protective gloves

### Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P308 + P313 - IF exposed or concerned: Get medical advice/attention

P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 - Store in a well-ventilated place

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Toxicity to Soil Dwelling Organisms

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component  | CAS No    | Weight % |
|------------|-----------|----------|
| Nickel     | 7440-02-0 | 77.0     |
| Iron       | 7439-89-6 | 14.0     |
| Copper     | 7440-50-8 | 5.0      |
| Molybdenum | 7439-98-7 | 4        |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

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May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

approved class D extinguishers. Do not use water or foam.

#### Extinguishing media which must not be used for safety reasons

Water may be ineffective.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Nickel oxides, Molybdenum oxides, Copper oxides, Iron oxides.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. No special precautions required.

### Environmental precautions

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Pick up and transfer to properly labelled containers.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

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## Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry place. Keep away from acids.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component  | Malaysia | ACGIH TLV                                             | OSHA PEL                                                                                       |
|------------|----------|-------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Nickel     |          | TWA: 1.5 mg/m <sup>3</sup>                            | (Vacated) TWA: 1 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup>                                 |
| Copper     |          | TWA: 0.2 mg/m <sup>3</sup>                            | (Vacated) TWA: 0.1 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> |
| Molybdenum |          | TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> | (Vacated) TWA: 10 mg/m <sup>3</sup>                                                            |

| Component  | European Union | The United Kingdom                                                                                                                         | Germany                                                                                                                                     |
|------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Nickel     |                | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr<br>Skin                                                              | TWA: 0.03 mg/m <sup>3</sup> (8 Stunden).<br>AGW - exposure factor 8<br>TWA: 0.006 mg/m <sup>3</sup> (8 Stunden).<br>AGW - exposure factor 8 |
| Copper     |                | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr | TWA: 0.01 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.02 mg/m <sup>3</sup>                                                           |
| Molybdenum |                | STEL: 20 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr                                                                        |                                                                                                                                             |

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

#### Hand Protection

No special protective equipment required

#### Skin and body protection

Long sleeved clothing

#### Respiratory Protection

No special protective equipment required

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

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## Information on basic physical and chemical properties

|                                         |                            |                                   |
|-----------------------------------------|----------------------------|-----------------------------------|
| Appearance                              |                            |                                   |
| Physical State                          | Solid Foil                 |                                   |
| Odor                                    | No information available   |                                   |
| Odor Threshold                          | No data available          |                                   |
| pH                                      | No information available   |                                   |
| Melting Point/Range                     | No data available          |                                   |
| Softening Point                         | No data available          |                                   |
| Boiling Point/Range                     | No information available   |                                   |
| Flash Point                             | No information available   | Method - No information available |
| Evaporation Rate                        | Not applicable             | Solid                             |
| Flammability (solid,gas)                | No information available   |                                   |
| Explosion Limits                        | No data available          |                                   |
| Vapor Pressure                          | No data available          |                                   |
| Vapor Density                           | Not applicable             | Solid                             |
| Specific Gravity / Density              | No data available          |                                   |
| Bulk Density                            | No data available          |                                   |
| Water Solubility                        | Insoluble in water         |                                   |
| Solubility in other solvents            | No information available   |                                   |
| Partition Coefficient (n-octanol/water) |                            |                                   |
| Autoignition Temperature                | No data available          |                                   |
| Decomposition Temperature               | No data available          |                                   |
| Viscosity                               | Not applicable             | Solid                             |
| Explosive Properties                    | No information available   |                                   |
| Oxidizing Properties                    | No information available   |                                   |
| Molecular Formula                       | Ni:Fe:Cu:Mo; 77:14:5:4 wt% |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

### Conditions to Avoid

None known.

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## Incompatible Materials

Acids.

## Hazardous Decomposition Products

Nickel oxides. Molybdenum oxides. Copper oxides. Iron oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

No data available

Inhalation

No data available

### Toxicology data for the components

| Component  | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation              |
|------------|---------------------------|---------------------------|------------------------------|
| Nickel     | LD50 > 9000 mg/kg ( Rat ) | -                         | LC50 > 10.2 mg/L ( Rat ) 1 h |
| Iron       | 7500 mg/kg ( Rat )        | -                         | -                            |
| Copper     | -                         | -                         | LC50 > 5.11 mg/L ( Rat ) 4 h |
| Molybdenum | -                         | LD50 > 2000 mg/kg ( Rat ) | LC50 > 5.84 mg/L ( Rat ) 4 h |

##### (b) skin corrosion/irritation;

No data available

##### (c) serious eye damage/irritation;

No data available

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

Category 1

May cause sensitization by skin contact

##### (e) germ cell mutagenicity;

No data available

##### (f) carcinogenicity;

Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU | UK | Germany | IARC     |
|-----------|----|----|---------|----------|
| Nickel    |    |    | Cat. 1  | Group 2B |

##### (g) reproductive toxicity;

No data available

##### (h) STOT-single exposure;

No data available

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(i) STOT-repeated exposure; Category 1

Route of exposure Inhalation  
Target Organs Lungs.

(j) aspiration hazard; Not applicable  
Solid

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment. Contains a substance which is: Very toxic to aquatic organisms. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Water Flea                                    | Freshwater Algae                                                                                                                                   | Microtox |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Nickel    | LC50: > 100 mg/L, 96h (Brachydanio rerio)<br>LC50: = 1.3 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: = 10.4 mg/L, 96h static (Cyprinus carpio)                                                                                                                                                                                                                                                                                                                 | EC50 = 510 µg/L 96h                           | EC50 = 0.1 mg/L 72h<br>EC50 = 0.18 mg/L 72h                                                                                                        |          |
| Copper    | LC50: = 1.25 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 0.3 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: = 0.8 mg/L, 96h static (Cyprinus carpio)<br>LC50: = 0.112 mg/L, 96h flow-through (Poecilia reticulata)<br>LC50: = 0.052 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.0068 - 0.0156 mg/L, 96h (Pimephales promelas)<br>LC50: < 0.3 mg/L, 96h static (Pimephales promelas)<br>LC50: = 0.2 mg/L, 96h flow-through (Pimephales promelas) | EC50: = 0.03 mg/L, 48h Static (Daphnia magna) | EC50: 0.031 - 0.054 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: 0.0426 - 0.0535 mg/L, 72h static (Pseudokirchneriella subcapitata) |          |

**Persistence and degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
**Persistence** Insoluble in water, May persist.  
**Degradability** Not relevant for inorganic substances.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

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|                                        |                                                                                                            |
|----------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Bioaccumulative potential</b>       | May have some potential to bioaccumulate; Product has a high potential to bioconcentrate                   |
| <b>Mobility in soil</b>                | Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility. |
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors                                  |
| <b>Other adverse effects</b>           | No information available                                                                                   |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste treatment methods</b>             |                                                                                                                                                                     |
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                           |
| <b>Other Information</b>                   | Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains                   |

## SECTION 14: TRANSPORT INFORMATION

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>IMDG/IMO</b>                     | Not regulated                   |
| <b>Road and Rail Transport</b>      | Not regulated                   |
| <b>IATA</b>                         | Not regulated                   |
| <b>Special Precautions for User</b> | No special precautions required |

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component  | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Nickel     | 231-111-4 | X    | X   | X     | X    |      | X     | X    | KE-25818 |
| Iron       | 231-096-4 | X    | X   | X     | X    |      | X     | X    | KE-21059 |
| Copper     | 231-159-6 | X    | X   | X     | X    |      | X     | X    | KE-08896 |
| Molybdenum | 231-107-2 | X    | X   | X     | X    |      | X     | X    | KE-25427 |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance



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Ozone Depletion Potential

This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Prepared By

Health, Safety and Environmental Department

Revision Date

31-Mar-2025

Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**